PRODUCT REQUIREMENTS DOCUMENT

CareCompanion AI

Smart Elderly Independence & Safety Platform

AI-Driven Healthcare Hackathon 2025

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1. Executive Summary

CareCompanion AI is an intelligent, voice-first healthcare companion platform designed specifically for elderly individuals who wish to maintain their independence, safety, and well-being at home. Leveraging cutting-edge artificial intelligence, the platform delivers real-time health monitoring, proactive medication management, personalized nutrition guidance, and seamless connectivity with caregivers and medical professionals.

This document defines the complete product requirements for CareCompanion AI, encompassing all features, user stories, technical specifications, success metrics, and implementation roadmap for the AI-Driven Healthcare Hackathon 2025.

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Global Elderly Population	Prefer Independent Living	Medication Non-Adherence	Emergency Response Gap

2. Problem Statement

The global elderly population faces a compounding crisis of health management, social isolation, and safety risks. As individuals age, managing daily health routines becomes increasingly complex, while access to timely medical support remains limited. Families and caregivers often live at a distance, creating critical gaps in oversight and emergency response.

2.1 Core Challenges

- Medication non-adherence affects over 50% of elderly patients, leading to hospitalizations and preventable health complications.
- Delayed emergency response — on average 15 minutes — can be life-threatening in falls, strokes, or cardiac events.
- Nutritional deficiencies are common due to lack of personalized dietary guidance tailored to age-related conditions.
- Social isolation accelerates cognitive decline and mental health deterioration among seniors living alone.
- Language and technology barriers prevent many elderly users from accessing digital health tools.
- Caregivers and physicians lack real-time visibility into the daily health status of elderly patients between appointments.

3. Product Vision & Goals

VISION	To be the most trusted AI companion for every elderly individual — empowering them to live independently, safely, and with dignity through intelligent, compassionate technology.

3.1 Strategic Goals

1. Enhance Independence — Enable elderly users to confidently manage their own health without constant caregiver intervention.
2. Ensure Safety — Provide real-time monitoring and instant emergency response capabilities.
3. Improve Health Outcomes — Reduce medication errors, improve nutrition, and enable proactive care.
4. Bridge the Care Gap — Connect patients, families, and doctors through a unified, real-time platform.
5. Break Language Barriers — Support multilingual voice interaction to serve diverse elderly populations.

4. Target Users & Personas

4.1 Primary Users

Persona	Description	Key Needs

Rajan, 73 — Retired Teacher	Lives alone, manages diabetes, uses smartphone occasionally. Primary language: Kannada.	Voice reminders, diet tips, emergency alert, doctor calls
Dr. Priya, 42 — Cardiologist	Monitors 30+ elderly patients remotely, needs a dashboard for vitals and activity.	Patient dashboard, analytics, quick video consults
Meera, 45 — Daughter/Caregiver	Lives in another city, worried about her father's medication adherence and safety.	Real-time location, activity alerts, emergency notifications

5. Feature Requirements

The following section defines all product features in detail, organized by functional module. Each feature includes a unique ID, description, user stories, acceptance criteria, and priority classification.

5.1 AI Voice Chatbot with Multilingual Support

MODULE	Voice Interaction Engine — Multilingual AI Companion

The AI chatbot serves as the primary interface for elderly users, enabling natural, conversational voice interaction without requiring technical literacy. The system supports multiple regional and international languages, making it accessible to diverse elderly populations.

User Stories

- As an elderly user, I want to speak naturally in my mother tongue so that I can interact with the app without language barriers.
- As a user with limited eyesight, I want to receive audio responses so that I don't have to read text on the screen.
- As a user, I want the AI to understand context across conversations so that I don't repeat myself.

Functional Requirements

- Voice-to-text (STT) and text-to-voice (TTS) in a minimum of 10 languages including English, Hindi, Kannada, Tamil, Telugu, Bengali, Marathi, Gujarati, Malayalam, and Punjabi.
- Real-time language detection — automatically identify user's preferred language based on spoken input.
- Conversational context memory — maintain conversation history within a session for natural follow-ups.
- Wake word activation (e.g., 'Hey CareCompanion') for hands-free access.
- Fallback to text input when voice recognition confidence is below threshold.

- Large-font, high-contrast UI text alongside all voice outputs for accessibility compliance.
- Noise cancellation and elderly speech pattern optimization (slower pace, varied pronunciation).

Acceptance Criteria

6. System correctly identifies spoken language with >90% accuracy across all supported languages.
7. Voice recognition accuracy for elderly speech patterns achieves >85% on standard test sets.
8. Response latency for voice queries is under 3 seconds on standard 4G connection.
9. System handles ambient noise up to 60dB without significant accuracy degradation.

5.2 Daily Medication & Activity Reminders

MODULE	Smart Reminder & Scheduling Engine

The reminder system provides personalized, multimodal alerts for medications, exercises, doctor appointments, hydration, and other daily health tasks. Reminders are delivered via voice, push notification, and SMS.

User Stories

- As an elderly user, I want to receive a voice reminder before my medication time so that I never miss a dose.
- As a caregiver, I want to receive a notification if my parent skips a medication so that I can follow up.
- As a user, I want to add reminders by speaking to the AI so that I don't have to type anything.

Functional Requirements

- Configurable reminder types: medications, meals, hydration, exercises, doctor appointments, and custom tasks.
- Multi-channel delivery: in-app voice, push notification, SMS, and WhatsApp integration.
- Confirmation mechanism — user must confirm reminder completion via voice ('Done') or button tap.
- Escalation protocol — unconfirmed critical reminders (medications) trigger caregiver notification after 15 minutes.
- Smart scheduling — AI suggests optimal medication timing based on user routine and past behavior.
- Medication photo recognition — user can photograph a pill bottle to auto-populate medication details.
- Recurring reminder patterns: daily, weekly, specific days, as-needed.

Acceptance Criteria

10. Reminder delivery success rate >99% across all channels.
11. User can add a new medication reminder in under 60 seconds via voice.
12. Caregiver escalation alert sent within 2 minutes of missed confirmation.

5.3 AI-Powered Diet & Nutrition Recommendations

MODULE	Personalized Nutrition Intelligence Engine

The nutrition module provides personalized, culturally relevant dietary guidance tailored to each user's age, health conditions (diabetes, hypertension, cardiac issues), medications, and regional food preferences.

Functional Requirements

- Health profile integration — diet recommendations adapt to diagnosed conditions, allergies, and medications.
- Cultural and regional food database covering Indian regional cuisines, international diets, and dietary restrictions (vegetarian, vegan, halal, jain).
- Daily meal planning with portion guidance and nutritional breakdown (calories, protein, carbs, fiber, sodium).
- Food interaction alerts — warn if a food item conflicts with a current medication (e.g., grapefruit with statins).
- Hydration tracking and reminders based on user weight, climate, and medication type.
- Voice-queried food logging ('I just had idli and sambar') with automatic nutritional calculation.
- Weekly nutrition summary and trend analysis in the caregiver/doctor dashboard.

5.4 Activity & Wellness Monitoring

MODULE	AI Activity Detection & Wellness Tracker

The activity monitoring module detects whether the user is active, sedentary, or potentially in distress by analyzing interaction patterns, movement data (via paired wearable or phone sensors), and daily routine baselines.

Functional Requirements

- Passive activity inference from phone motion sensors, app interactions, and check-ins.
- Optional wearable integration (Google Fit, Apple Health, Fitbit, Samsung Health) for step count, heart rate, and sleep data.
- Baseline learning — AI establishes a personal activity baseline in the first 7 days of use.
- Inactivity alert — if no interaction is detected for a configurable period (default: 4 hours), alert caregiver and trigger a check-in call.

- Daily wellness check-in: AI asks the user 3 simple questions each morning about sleep, pain levels, and mood.
- Anomaly detection — identify deviations from routine (not waking at usual time, no kitchen activity, etc.).
- Activity trend reporting in caregiver and doctor dashboards.

5.5 Emergency Contacts & SOS Alert System

MODULE	Emergency Response System

The emergency system provides instant, multi-channel alerts to designated contacts and emergency services. It supports both manual SOS triggers and AI-detected emergency situations.

Functional Requirements

- Configurable emergency contact list — up to 5 contacts with priority ordering.
- One-tap and voice-activated SOS ('Help!' or 'Emergency') that triggers simultaneous calls and messages.
- SOS alert includes: user's name, current GPS location, health profile summary, and last known activity.
- Automatic integration with local emergency services (112 in India) with configurable toggle.
- False alarm cancellation — 10-second countdown with cancel option after accidental trigger.
- Fall detection — AI detects sudden phone drop/movement pattern and prompts user; escalates if no response within 30 seconds.
- Emergency contact dashboard showing alert history, response time, and resolution status.

Acceptance Criteria

13. SOS alert delivered to all emergency contacts within 30 seconds of trigger.
14. False alarm rate for automated fall detection is below 5%.
15. System operates in low-connectivity mode (SMS fallback) when internet is unavailable.

5.6 Live Location Sharing

MODULE	Real-Time GPS Location Service

The location module enables designated caregivers to view the elderly user's real-time location on a map, with configurable safe zone alerts and location history.

Functional Requirements

- Real-time GPS tracking with 30-second refresh intervals (configurable to battery-saving mode at 5-minute intervals).

- Safe zone configuration — caregivers define geographic boundaries; alerts triggered when user leaves or enters zones.
- Location history — 30-day location log accessible to authorized caregivers and doctors.
- Low battery alert — notify caregiver when user's device battery drops below 20%.
- Offline location — last known GPS coordinates displayed when device is offline.
- User privacy controls — user can pause location sharing with a voice command or button; caregiver receives a notification.
- Google Maps integration for route visualization and navigation assistance.

5.7 Video Consultation with Doctors

MODULE	Telehealth Video Call Platform

The telehealth module enables secure, high-quality video consultations between elderly users and their registered physicians directly within the app, eliminating the need for third-party video call applications.

Functional Requirements

- In-app video calling using WebRTC with end-to-end encryption (HIPAA/DPDP compliant).
- Appointment scheduling — users and doctors can schedule, reschedule, or cancel appointments.
- Pre-call health summary — automatically compiled from recent vitals, medications, and activity data, shared with the doctor at call start.
- One-tap call initiation optimized for elderly users — large buttons, clear audio, bright interface.
- Call recording with consent — session recordings stored securely and accessible to patient and doctor.
- Post-call notes — doctors can add prescription updates, notes, and follow-up reminders directly in the app.
- Connection quality optimization — adaptive video resolution based on network conditions.

5.8 User & Doctor Authentication System

MODULE	Identity Management & Role-Based Access

The authentication module provides secure, role-based access for elderly users, caregivers, and medical professionals, with elderly-friendly login options.

Functional Requirements

- Three role types: Elderly User, Caregiver/Family, and Medical Professional (Doctor).

- Elderly user login options: 4-digit PIN, biometric (fingerprint/face ID), or OTP via SMS — no password required.
- Doctor verification: medical license number validation, document upload, and manual approval workflow.
- Caregiver linking: QR code or invite link sent by the elderly user to authorize caregiver access.
- Role-based permissions: doctors see clinical data; caregivers see activity, location, and alerts; users see personal dashboard.
- Session management: auto-logout after 30 minutes of inactivity for security.
- Multi-factor authentication for doctor accounts accessing sensitive patient data.

5.9 Analytics Dashboard & Reporting

MODULE	Health Intelligence Dashboard

The analytics dashboard provides caregivers and doctors with a comprehensive, data-rich view of the elderly user's health trends, activity patterns, medication adherence, and wellness indicators.

Dashboard Modules — Caregiver View

- Activity Heatmap — visual daily activity pattern over the past 30 days.
- Medication Adherence Rate — percentage compliance chart with missed dose log.
- Emergency Alert History — timeline of SOS events, fall detections, and responses.
- Nutrition Compliance — weekly diet adherence vs. recommended targets.
- Location History Map — 30-day GPS trail with timeline.
- Wellness Trend — mood, sleep, and pain level trends from daily check-ins.

Dashboard Modules — Doctor View

- Patient Health Summary — consolidated view of all elderly patients under care.
- Vitals Trend Charts — heart rate, blood pressure, glucose levels (from wearable/manual input).
- Medication Interaction Flags — AI-identified potential drug interaction alerts.
- Appointment Calendar — upcoming consultations with pre-call health summaries.
- Report Export — PDF/CSV export of any patient's health data for records.

Acceptance Criteria

16. Dashboard loads within 2 seconds for up to 90 days of historical data.
17. All charts update in real-time when new data is received.
18. PDF report generation completes in under 10 seconds.

6. Feature Priority Matrix

The following table summarizes all platform features with priority classification and implementation effort estimates.

Feature ID	Feature Name	Description	Priority	Effort
F-001	AI Voice Chatbot	Multilingual voice interaction, STT/TTS, context memory, wake word	P0 - Critical	High
F-002	Medication Reminders	Smart reminders, multi-channel delivery, caregiver escalation	P0 - Critical	Medium
F-003	Emergency SOS System	One-tap SOS, fall detection, multi-contact alerts, GPS in alert	P0 - Critical	Medium
F-004	User/Doctor Login	Role-based access, PIN/biometric login, doctor verification	P0 - Critical	Medium
F-005	Live Location Sharing	Real-time GPS, safe zones, location history, caregiver alerts	P1 - High	Medium
F-006	Activity Monitoring	Inactivity detection, wellness check-in, anomaly alerts	P1 - High	High
F-007	Video Doctor Calls	In-app WebRTC video, scheduling, call notes, health summary sharing	P1 - High	High
F-008	Diet & Nutrition AI	Personalized meal plans, food interactions, voice food logging	P1 - High	Medium
F-009	Analytics Dashboard	Health trends, adherence reports, doctor and caregiver views	P2 - Medium	Medium
F-010	Wearable Integration	Google Fit, Apple Health, Fitbit, Samsung Health sync	P2 - Medium	Medium

7. Technical Architecture

7.1 System Architecture Overview

CareCompanion AI is built on a cloud-native, microservices architecture ensuring scalability, resilience, and HIPAA/DPDP compliance. Each module operates as an independent service communicating via a secure API gateway.

Layer	Technology Stack
Mobile App	React Native (iOS & Android), Expo SDK, NativeWind for styling
Voice AI	Google Cloud Speech-to-Text / Azure Cognitive Services, Whisper (offline), TTS via Google WaveNet

AI/ML Backend	Python (FastAPI), LangChain, OpenAI GPT-4o for conversational AI, scikit-learn for activity anomaly detection
Video Calls	WebRTC with Twilio Video / 100ms.live SDK, AES-256 encryption
Backend API	Node.js (Express) + Python microservices, REST + WebSocket for real-time features
Database	PostgreSQL (user/health data), Redis (session/cache), MongoDB (activity logs), InfluxDB (time-series vitals)
Cloud	AWS (ECS, RDS, S3, CloudFront) or GCP — multi-AZ deployment
Location	Google Maps Platform (Maps SDK, Geocoding, Places API)
Notifications	Firebase Cloud Messaging (push), Twilio (SMS/WhatsApp), AWS SES (email)
Security	OAuth 2.0 + JWT, TLS 1.3, AES-256 at rest, DPDP/HIPAA compliance layer

8. Non-Functional Requirements

8.1 Accessibility

- WCAG 2.1 AA compliance for all visual interfaces.
- Minimum font size of 18pt throughout the application.
- High-contrast mode and adjustable text size.
- Screen reader compatibility (TalkBack / VoiceOver).
- All interactions completable via voice — no mandatory touch input.

8.2 Performance

- App cold start time under 3 seconds on mid-range Android device.
- API response time under 500ms for 95th percentile of requests.
- Video call quality maintained at minimum 480p with <150ms latency.
- Offline capability — core reminders and emergency features functional without internet.

8.3 Security & Privacy

- All health data encrypted at rest (AES-256) and in transit (TLS 1.3).
- Compliance with India's Digital Personal Data Protection Act (DPDP) 2023.
- No third-party advertising SDK or data sharing.
- Data residency in India — user health data stored on Indian servers.
- Right to erasure — users can request full data deletion within 30 days.

8.4 Reliability

- System uptime SLA of 99.9% (less than 9 hours downtime per year).

- Emergency features (SOS, location) maintain 99.99% availability via redundant architecture.
- Automatic failover for critical services with recovery time under 60 seconds.

9. Success Metrics & KPIs

The following key performance indicators define success for CareCompanion AI across user adoption, health outcomes, and platform reliability.

Metric	Target	Measurement Method
Medication Adherence Rate	>85% (from baseline ~50%)	Reminder confirmation logs
Daily Active Users (DAU)	>70% of registered users	App analytics
Emergency Response Time	<30 seconds for SOS delivery	Alert delivery timestamp logs
Voice Recognition Accuracy	>85% for elderly speech	STT evaluation benchmark
User Satisfaction (NPS)	>60 Net Promoter Score	In-app monthly survey
Doctor Engagement Rate	>80% complete patient reviews weekly	Dashboard login analytics
Caregiver Alert Response	<5 minutes average response time	Alert response tracking
App Crash Rate	<0.1% of sessions	Firebase Crashlytics
System Uptime	>99.9%	AWS CloudWatch monitoring

10. Implementation Roadmap

Phase	Timeline	Deliverables
Phase 1 (MVP)	Weeks 1-6	User & doctor login, AI voice chatbot (English + 2 languages), medication reminders, emergency SOS, live location sharing
Phase 2 (Core)	Weeks 7-12	Full multilingual support (10 languages), video doctor calls, activity monitoring, diet recommendations, caregiver dashboard
Phase 3 (Analytics)	Weeks 13-18	Full analytics dashboard, wearable integration, advanced anomaly detection, PDF report generation, medication photo recognition
Phase 4 (Scale)	Weeks 19-24	Performance optimization, security audit, DPDP compliance review, regional language expansion, WhatsApp integration

11. Risks & Mitigations

Risk	Severity	Mitigation Strategy

Low voice recognition accuracy for elderly speech patterns	**High**	Train on elderly speech datasets; use Whisper fine-tuned model; implement manual correction fallback.
Privacy concerns around health data and location tracking	**High**	Transparent consent UI; granular privacy controls; DPDP compliance; no data sharing with third parties.
Elderly user resistance to technology adoption	**Medium**	Caregiver-assisted onboarding; in-home tutorial mode; 24/7 voice support helpline.
Network connectivity issues in rural areas	**Medium**	Offline-first architecture for core features; SMS fallback for alerts; data compression for low-bandwidth.
False positives in emergency detection causing alert fatigue	**Medium**	10-second cancel window; configurable sensitivity; machine learning calibration per user.
Regulatory approval for telehealth features	**Medium**	Engage with MoHFW guidelines; obtain Telemedicine Practice Guidelines compliance certification.

12. Conclusion

CareCompanion AI represents a comprehensive, compassionate, and technically robust solution to one of society's most pressing challenges — enabling elderly individuals to live independently, safely, and with dignity.

By combining AI-powered voice interaction, real-time health monitoring, seamless caregiver connectivity, and proactive medical support, the platform addresses the full spectrum of elderly care needs. The multilingual approach ensures inclusivity across India's diverse linguistic landscape, while the role-based architecture serves patients, families, and medical professionals with equal effectiveness.

HACKATHON IMPACT	CareCompanion AI has the potential to transform elderly care for over 140 million elderly Indians — reducing medication errors, cutting emergency response times, improving nutritional health, and giving families peace of mind through intelligent, always-available AI companionship.

Document Prepared By:	**Review & Approval:**
Product & Engineering Team CareCompanion AI — Hackathon Team	Version 1.0 — Pending Hackathon Review *Confidential — Do Not Distribute*